

Presentation 2

- Links to presentation(s) and code(s) on GitHub

Code:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Chatbot%20analysis/ChatbotAnalysisCode_20Oct_Marcos.ipynb

Presentation:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Chatbot%20analysis/ChatbotAnalysis_20Oct_Marcos.pptx

- What did you do?

I completed tasks relating to analysis of the chatbot. I redid the analysis that I did last time with some tweaks. I analyzed how termination reasons look different for calls that reached the chatbot vs some that did not (I needed to redo this because I got a large number of N/A's as termination reasons in my first analysis). I also analyzed the duration of calls before, during, and after chatbot usage using a boxplot, which gave me a better idea of the distribution of duration within each period. Finally I looked at what people spent their time on within the family menu and how that looked different in the different periods i described above.

- How does it help the project?

It helps the project by getting a better understanding of the chatbot's impact on the customer experience and ultimately will help inform the decision to purchase or not purchase it moving forward

- Issues faced (if any)

The CAR data does not have a specific column specifying how long someone was doing an activity for so seeing what people spent time on was more difficult.

- Attempts to resolve issues (if any)

In order to solve this, I tracked the contact session ID and tracked the time between when someone started doing an activity and when it was logged that they started doing something else.

- Issues resolved (if any)

- Next steps

In the termination reason analysis I will limit the control group to calls that reached a human agent or compare termination reasons during the broad chatbot period vs before/after the chatbot period because at this point we might be comparing apples to oranges. The reason why agent left termination reason for the chatbot reached calls is so much higher than those that didn't may be because most calls just don't reach an agent at all

- References (Mention if you built up on someone else's work)

Presentation 1

- Links to presentation(s) and code(s) on GitHub

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/tree/main/Code/Chatbot%20analysis
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/tree/main/Presentations/Chatbot%20analysis

- What did you do?

I did the chatbot analysis part of the presentation. More specifically I looked to do an analysis on the termination reasons when chatbots were and were not used post-March (when the termination column was added) and a duration of calls analysis during the time period that the chatbot was in use vs. when it was not. The termination reason analysis was meant to provide insight on if the call was successful in helping the client or not, and the call duration analysis was meant to analyze efficiency with and without the chatbot

- How does it help the project?

This helps the project because it helps get us closer to answering the question of the effectiveness of the chatbot. This is an important business decision that Cynthia wants to make. Is it worth purchasing this chatbot permanently? This helps answer that question. The insight that we got from the analysis is that when we remove calls with an

empty termination reason, it seems that chatbot calls are more efficient because an agent ended the call a higher percentage of the time, which we are assuming means that the issue was more likely to have been resolved. We also saw a much higher duration from the calls that had encountered the chatbot, which can be potentially inefficientness, but more analysis needs to be done to determine the reason for this.

- Issues faced (if any)

There were a lot of missing values for termination reasons amongst the contact session ID's. This can possibly impact the proportion I get when I removed these N/A's and lead to inaccurate representations of the real data and as a result, incorrect conclusions.

- Attempts to resolve issues (if any)

I tried finding out why these calls with N/As, if there was something that was different about those calls, but I couldn't find anything.

- Issues resolved (if any)

- Next steps

I will try to see if there is another way that I can possibly think about measuring outcome effectiveness. I will also shorten my termination analysis specifically to august inside of going back to March to see if I can get values for all calls during the month and compare this to the proportion that I currently have. I also would like to dig into why the chatbot calls might have had such a long duration.

- References (Mention if you built up on someone else's work)